

Solving inequalities

Inequalities can be solved by rearranging them, but anything that is done to one side of the inequality must also be done to the other. For example, any number added to cancel a numerical term from one side must be added to the numerical term on the other side.

To solve this inequality,
add 2 to both sides then divide
by 3.

$$3b - 2 \geq 10$$

To isolate 3b, -2 needs to be
removed, which means adding
+2 to both sides.

adding 2 to $3b - 2$
leaves $3b$ on its own

$$3b \geq 12$$

$$10 + 2 = 12$$

Solve the inequality by dividing
both sides by 3 to isolate b .

$3b$ divided by 3
leaves b on its own

$$b \geq 4$$

$$12 \div 3 = 4$$

To solve this inequality,
subtract 3 from both sides then
divide by 3.

$$3a + 3 < 12$$

Rearrange the inequality by
subtracting 3 from each side to
isolate the a term on the left.

subtracting 3 leaves
 $3a$ on its own

$$3a < 9$$

$$12 - 3 = 9$$

Solve the inequality by dividing
both sides by 3 to isolate a . This is
the solution to the inequality.

$3a$ divided
by 3 leaves a
on its own

$$a < 3$$

$$9 \div 3 = 3$$

Solving double inequalities

To solve a double inequality, deal with each side
separately to simplify it, then combine the two sides
back together again in a single answer.

This is a double inequality that needs
to be split into two smaller inequalities
for the solution to be found.

$$-1 \leq 3x + 5 < 11$$

These are the two parts the double
inequality is split into; each one needs
to be solved separately.

$$-1 \leq 3x + 5$$

$$3x + 5 < 11$$

subtracting 5 from
-1 gives -6

$$-6 \leq 3x$$

subtracting 5
from $3x + 5$
leaves $3x$ on
its own

Isolate the x terms by subtracting
5 from both sides of the smaller parts.

subtracting 5
from $3x + 5$ leaves
 $3x$ on its own

$$3x < 6$$

subtracting
5 from 11
gives 6

$$-6 + 3 = -2$$

$$-2 \leq x$$

$$3x \div 3 = x$$

Solve the part inequalities by
dividing both of them by 3.

$$3x \div 3 = x$$

$$x < 2$$

$$6 \div 3 = 2$$

$$-2 \leq x < 2$$

**Finally, combine the two small
inequalities** back into a single
double inequality, with each in the
same position as it was in the original
double inequality.